Investigating Phase Limitations in STFT-Based Speech Enhancement

A controlled oracle-phase analysis

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Abstract

Classical speech enhancement commonly reconstructs an estimated magnitude spectrum with the original noisy phase. This project measures the cost of that choice by swapping in clean reference phase while holding the magnitude estimate fixed. Spectral subtraction and Wiener filtering were evaluated on 360 mixtures built from 40 VoiceBank utterances, three noise conditions and three input SNRs. The mean phase swap added 0.0170 STOI, 0.0661 PESQ and 1.96 dB SI-SDR for spectral subtraction. The association between phase disagreement and STOI gain remained positive after accounting descriptively for noise and SNR. The negative pooled PESQ association largely disappeared for spectral subtraction after the same adjustment. An exact-magnitude diagnostic and a pretrained MP-SENet comparison on 167 matched mixtures put these changes in context. The results support phase correction as a targeted intervention and identify magnitude estimation as a separate source of error.

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Project materials

GitHub Project Site · Code repository · Demonstration video

Research question and scope

Single-channel speech enhancement estimates a clean signal from a recording that also contains noise. In the short-time Fourier transform (STFT), a time frame is represented by complex coefficients. Their magnitudes describe the strength of components and their phases describe the alignment of those components within the signal. Magnitude estimators such as spectral subtraction and Wiener filtering can produce a new magnitude without producing a new phase. Reusing the mixture phase then makes waveform reconstruction simple, but assumes its errors are tolerable.

An ordinary comparison of two enhancement systems cannot tell how much of a performance difference is due to their phase handling. Their gains, spectral estimates and model capacity may differ simultaneously. The original proposal used two classical methods as a starting point. The revised experiment turns each into its own control by supplying two phases to the same magnitude estimate. The final analysis asks where that controlled change helps, where it fails to help, and which observations survive stratification by noise and SNR.

Research question

How much does reusing noisy phase limit classical STFT speech enhancement across noise conditions and input SNRs when the estimated magnitude is held fixed?

Scope and contribution

The phase intervention is evaluated at 0, 5 and 10 dB using cafeteria, traffic and generated white noise. Both classical baselines use the known noise component to estimate a frequency-dependent average noise spectrum. That access is available in the controlled mixture construction. Deployment on a single noisy recording would require estimating the noise spectrum from available input.

The project combines two established estimators, a paired phase substitution, a clean-magnitude diagnostic, and a matched comparison with a published pretrained model. The final contribution is the paired analysis by noise and SNR, including counts of individual regressions and a check on the interpretation of pooled phase-disagreement correlations. MP-SENet supplies context for the attainable scores of a different system.

Reading the results

Three objective measures describe different properties of the output. STOI estimates intelligibility, wideband PESQ estimates perceptual quality, and SI-SDR measures scale-invariant waveform fidelity. A change can move these measures in different directions. The analysis therefore reports three separate effects and treats the listening samples as an illustration alongside the objective measures.

Why isolate phase?

Magnitude-domain estimators

Boll [1] estimates noise spectra for subtraction, offering a transparent way to attenuate spectral components associated with noise. The implementation here subtracts the mean noise magnitude per frequency and applies a floor. This limits excessive attenuation but can leave residual noise. Wiener filtering uses an estimate of signal power relative to total power to derive a smooth gain [2]. Here the speech-power estimate comes from a clipped difference between instantaneous mixture power and time-averaged noise power. Both estimators are inexpensive and suitable for the phase substitution because the magnitude is explicit.

Oracle reconstruction and phase sensitivity

Paliwal et al. [3] show that spectral phase can matter to enhancement and that its usefulness depends on the analysis and reconstruction setup. Their work motivates changing phase independently at the synthesis input. Clean phase is available here through the clean reference used to create the mixture. It provides a diagnostic intervention with known phase. A clean-phase substitution with an imperfect magnitude is one particular reconstruction and can produce worse metric scores for some utterances. It should be interpreted as the measured effect of this intervention.

Wrapped phase is circular: angles near opposite ends of the interval from minus pi to pi may be close on the circle. A wrapped absolute difference avoids treating this boundary as a large disagreement. Weighting by clean magnitude gives speech-bearing bins greater influence. A single weighted mean still discards where errors occur in time and frequency, which matters when the metrics weight speech structure differently.

Complex and explicitly phase-aware models

Complex ratio masking [4] allows the mask to rotate as well as rescale coefficients. DCCRN [5] applies complex-valued convolutional and recurrent operations to predict a complex representation. These approaches can change magnitude and phase together, but a system-level comparison leaves their individual contributions entangled. MP-SENet [6] has parallel magnitude and phase estimation paths and a published VoiceBank checkpoint. It is included as a phase-aware reference because it can run on the same noisy inputs without access to clean speech at inference. Its weights, architecture and training also differ from the classical methods.

Evaluation choices

Taal et al. [9] motivate STOI through short-time speech-envelope comparisons in frequency bands. PESQ [10] compares reference and degraded audio using a perceptual model. Le Roux et al. [11] define SI-SDR through projection onto the clean waveform. Those complementary targets motivate separate reporting. Even with an exact input magnitude, imposing a different phase and applying inverse STFT can change the waveform and the magnitude obtained if that waveform is analysed again.

Controlled mixture and reconstruction

Data and transform

The archive contains 40 VoiceBank utterances from speakers p232 and p257 [7]. Cafeteria and traffic recordings come from DEMAND [8]. Gaussian white noise was generated with seed 5305. The noise is length-matched to each clean utterance, then scaled by the square root of clean power divided by the product of noise power and 10 raised to SNR/10. Adding the scaled noise gives the mixture. The complete 40 × 3 × 3 grid contains 360 mixtures. At 16 kHz, the 512-sample Hann window spans 32 ms and the 256-sample hop spans 16 ms.

Magnitude estimates and controls

Spectral subtraction takes the maximum of the mixture magnitude minus mean noise magnitude and 0.02 times mixture magnitude. Wiener filtering first estimates speech power as mixture power minus mean noise power, clipped at zero. The gain is the estimated speech power divided by its sum with the noise power (plus a numerical stabiliser), with a minimum gain of 0.02. The mixture magnitude is multiplied by this gain.

Condition family	Synthesis magnitude	Synthesis phase
Unprocessed reference	Mixture	Mixture
Spectral subtraction pair	Same subtraction estimate	Mixture or clean
Wiener pair	Same Wiener estimate	Mixture or clean
Exact-magnitude diagnostic	Clean reference	Mixture
MP-SENet	Network estimate	Network estimate

Table 1. Conditions for the classical experiment and the matched neural comparison.

Each reconstruction forms a complex STFT from magnitude and phase and applies inverse STFT at the original signal length. Phase disagreement is the clean-magnitude-weighted mean absolute wrapped difference between the mixture and clean angles, in radians. Paired benefit is the clean-phase score minus the mixture-phase score for the same mixture and estimator.

Scores and analysis

Signals are trimmed to a common length. The experiment uses standard STOI, 16 kHz wideband PESQ, and SI-SDR computed from mean-centred signals. The paired tables use one record per mixture and condition. Pooled Pearson r uses all mixtures for each estimator. Cell-centred r subtracts each noise-by-SNR cell mean from both the disagreement and the gain before correlating residuals. This descriptive check removes between-cell mean differences. Each utterance recurs in nine cells, so the analysis makes no independent-sample significance claims.

The pretrained VoiceBank MP-SENet checkpoint was evaluated on the 167 mixtures processed within the available runtime. The classical scores are aligned by clean filename, noise type and input SNR. Each row of the matched table uses the same IDs. The final analysis checks the archived classical scores against the full dataset before comparing the matched means.

Paired effects across 360 mixtures

Estimator	Score	Mean gain	Median	Positive	Negative
SS	STOI	+0.0170	+0.0165	360	0
SS	PESQ	+0.0661	+0.0567	354	6
SS	SI-SDR (dB)	+1.9602	+2.0403	349	11
Wiener	STOI	+0.0164	+0.0156	360	0
Wiener	PESQ	+0.0608	+0.0488	354	6
Wiener	SI-SDR (dB)	+1.8978	+2.0308	345	15

Table 2. Clean-phase score minus noisy-phase score with the magnitude estimate held fixed. N = 360 per estimator and metric. There were no ties.

The intervention improved STOI in every pair in this archive. PESQ and SI-SDR rose on average but occasionally decreased, even with clean phase. The means therefore describe a typical benefit under these conditions, with exceptions that matter for interpreting the oracle as a diagnostic.

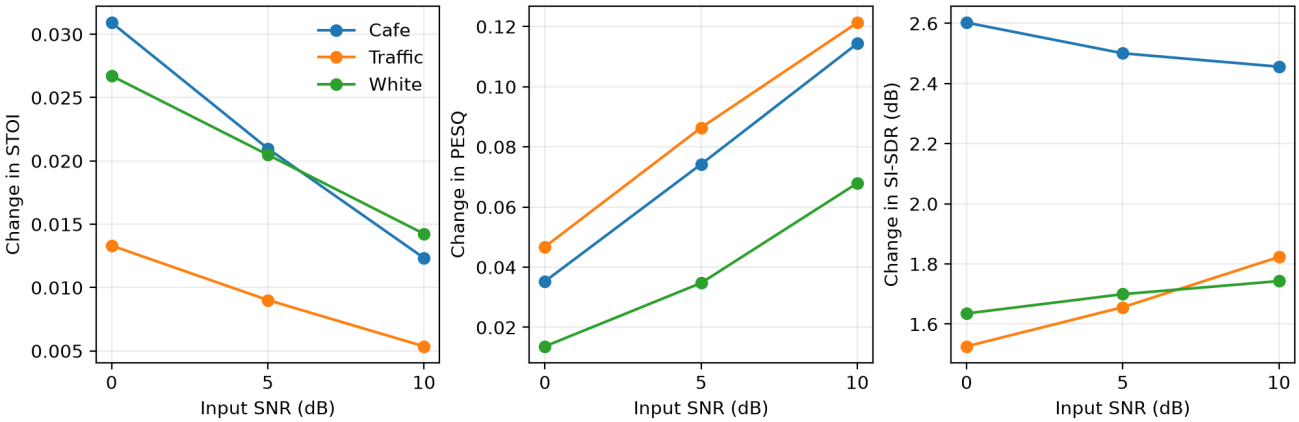


Figure 1. Mean change from the phase swap for spectral subtraction. Each point averages 40 mixtures at one noise type and input SNR. Panels use separate vertical scales.

For spectral subtraction in cafeteria noise, lowering input SNR from 10 to 0 dB increased mean STOI gain from 0.0123 to 0.0309. PESQ moved in the opposite direction, with gains of 0.1144 at 10 dB and 0.0353 at 0 dB. Mean SI-SDR gain was larger in cafeteria than in traffic or white noise at every tested SNR. The SNR trends for SI-SDR varied with noise type.

Reading changes in context

A gain is measured relative to the same estimator with noisy phase. For example, a positive STOI gain from the phase swap can coexist with little net improvement from the original noisy mixture. Across the full grid, mean noisy STOI was 0.8565, rising to 0.8593 with spectral subtraction and noisy phase and to 0.8763 with spectral subtraction and clean phase. The first contrast changes magnitude and the second changes phase.

The largest mean angle disagreement was in white noise at 0 dB (0.508 rad). The largest spectral-subtraction STOI gain was in cafeteria noise at 0 dB (0.0309). Comparing these cells shows why the average angle alone cannot rank the benefit across all conditions.

Phase disagreement and subgroup effects

A pooled correlation can reflect systematic differences across noise and SNR, variation among mixtures tested under the same conditions, or both. Comparing it with a cell-centred correlation helps separate these sources of association.

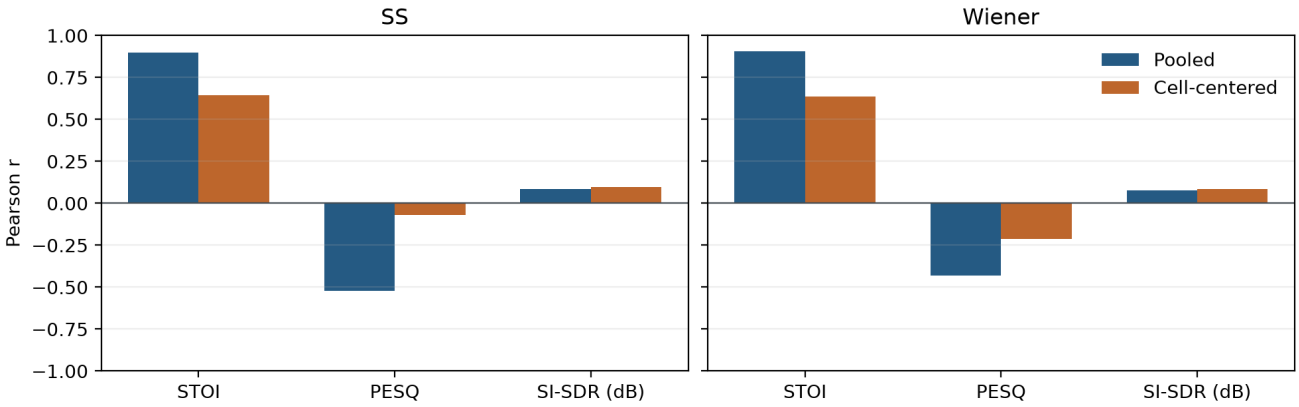


Figure 2. Pooled and cell-centred Pearson correlations between phase disagreement and the paired score gain. Each bar uses 360 observations for its estimator.

For spectral subtraction, STOI r fell from 0.897 to 0.643 after centring, and PESQ r moved from –0.523 to –0.068. Wiener STOI showed a similar residual association (0.635), while its PESQ association weakened from –0.429 to –0.213. SI-SDR r remained close to zero for both estimators despite a positive mean gain.

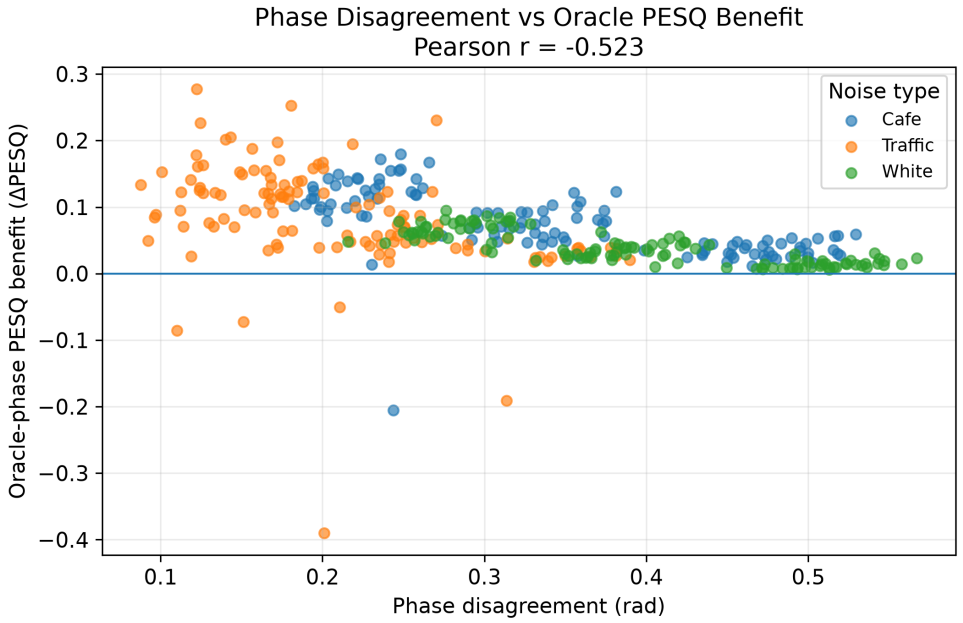


Figure 3. Archived spectral-subtraction PESQ scatterplot. The pooled relationship combines nine test cells and also includes negative gains.

Centring removes cell-level mean shifts but retains variation among utterances. For spectral subtraction, Figure 2 shows that the direction and strength of the STOI association differ from PESQ even after centring. This comparison does not remove other shared effects, such as speaker or utterance content. The angle also requires clean speech to compute, so it cannot directly select a phase-enhancement method from a noisy input alone.

Matched benchmark and acoustic example

Condition	STOI	PESQ	SI-SDR (dB)
Noisy	0.8595	1.2844	5.03
SS + noisy phase	0.8597	1.4639	9.16
SS + clean oracle phase	0.8749	1.5312	11.09
Wiener + noisy phase	0.8573	1.4098	8.53
Wiener + clean oracle phase	0.8720	1.4765	10.40
Perfect magnitude + noisy phase	0.9794	3.6227	15.13
MP-SENet	0.9231	2.3356	14.58

Table 3. Mean objective scores on 167 identical mixture IDs for each row. Exact clean magnitude and clean phase require access to the reference.

Compared with spectral subtraction using clean phase, MP-SENet added 0.0482 STOI, 0.8044 PESQ and 3.49 dB SI-SDR on the matched subset. It scored below the exact-magnitude diagnostic by 0.0562 STOI, 1.2871 PESQ and 0.55 dB SI-SDR. With noisy phase in both cases, replacing the spectral-subtraction magnitude by the true magnitude added 0.1197 STOI, 2.1589 PESQ and 5.97 dB SI-SDR. This quantifies the magnitude intervention under the same phase.

One cafeteria mixture at 0 dB

Condition	STOI	PESQ	SI-SDR (dB)
Noisy	0.7343	1.1941	-0.13
SS + noisy phase	0.7277	1.2226	2.23
SS + clean oracle phase	0.7512	1.2703	4.75
Wiener + noisy phase	0.7273	1.1957	2.15
Wiener + clean oracle phase	0.7481	1.2421	4.60
MP-SENet	0.8403	1.9166	11.03

Table 4. Archived acoustic example for p232_001. The model scores here come from the representative audio evaluation. The matched batch scores come from its separate CSV.

Spectral subtraction raised SI-SDR from −0.13 to 2.23 dB with noisy phase while STOI fell from 0.7343 to 0.7277. Substituting clean phase raised STOI to 0.7512 and SI-SDR to 4.75 dB. The example demonstrates how improved waveform fidelity can coexist with an intelligibility proxy that first worsens and then improves. MP-SENet gave 0.8403 STOI and 11.03 dB SI-SDR on this example.

Listen to the aligned example and inspect its spectrograms.

Interpreting the metric disagreement

Why STOI can benefit more at lower SNR

In a bin where the interference vector is comparable to the speech vector, adding the two can rotate the resulting coefficient substantially. This is more likely at low SNR, but global SNR alone does not determine the local interference-to-speech ratio. An inverse STFT combines overlapping frames, so changing phase can alter the reconstructed envelopes even when the supplied magnitude is fixed. The strong residual STOI association suggests that phase disagreement tracks some of this variation within noise/SNR cells. The analysis does not locate which bins produced the gain or establish actual listener intelligibility.

Why the pooled PESQ trend can reverse expectations

At higher SNR, the classical magnitude estimate may contain less noise-related error, leaving more speech structure for the phase substitution to preserve. At lower SNR, residual interference or excessive attenuation in that magnitude may constrain the PESQ gain. This mechanism is consistent with the cafeteria result and the large exact-magnitude gains. The study did not directly measure which artefacts PESQ penalised. The attenuation of r after centring shows that the large pooled negative association mainly describes differences across test conditions for spectral subtraction. It offers little evidence that a larger phase error within one condition causes a smaller PESQ benefit.

Why SI-SDR still improves when its correlation is small

SI-SDR compares waveform residual energy after scale alignment [11]. Removing a phase error can improve alignment, which is consistent with the mean gains near 2 dB. The mean absolute angle weights frequency bins by clean magnitude and loses their timing and pattern. It need not track the change in residual energy. Individual negative gains are possible when the imperfect magnitude and the new phase interact during overlap-add, or when noise removal and speech distortion trade off differently under a given metric.

What the modern benchmark establishes

MP-SENet outperformed the evaluated classical systems, including the classical reconstructions given clean phase, on matched inputs. That advantage measures a difference between complete systems. The model also changes the magnitude, and differs in training and architecture. An inference ablation using the model’s magnitude with noisy versus estimated phase would isolate the value of its phase path more directly. Lu et al. [6] report PESQ 3.50 on their standard benchmark. This study’s 2.34 comes from a smaller custom mixture grid at different test conditions, so the scores describe different evaluations.

How the result relates to prior work

The matched and paired results are consistent with Paliwal et al. [3] on the relevance of phase, while adding a direct comparison across three metrics for the selected Hann-window setup. The large exact-magnitude effect shows that the relative leverage of the two interventions depends on which magnitude and phase are available. Because the diagnostics use different reference information, their gains are conditional contrasts rather than additive components of the total error.

What the experiment answers

For the tested classical estimators, noisy-phase reuse limited mean performance by roughly 0.016–0.017 STOI, 0.061–0.066 PESQ and 1.90–1.96 dB SI-SDR relative to supplying clean phase at the same estimated magnitude. The mean STOI gain was largest at 0 dB within each noise type and remained associated with phase disagreement after cell centring. The analogous PESQ relationship was substantially weaker after centring. Thus the experiment answers the research question for this grid and this reconstruction configuration. The magnitude diagnostic and the pretrained reference locate additional headroom beyond changing the classical phase alone.

Limits and a follow-up experiment

Only two speakers and two recorded environments were evaluated. Repeated mixtures from each utterance share speech content, and each recorded noise represents one environment sample. These observations describe the archive and do not establish population-level effects. A repeat with more held-out speakers, independent noise excerpts and additional window durations would test how stable the metric trends are. Clustered resampling by utterance could quantify uncertainty conditional on the selected speakers once the design supports it.

The MP-SENet subset covers 167 of 360 mixtures and was determined by completion, with 18 unique utterances from p232 and nine from p257. Café, traffic and white noise account for 57, 59 and 51 mixtures. The 0, 5 and 10 dB counts are 56, 54 and 57. The matched and excluded sets already differ in mean noisy PESQ (1.2844 vs 1.1918) and spectral-subtraction SI-SDR (9.16 vs 9.86 dB). Matching controls which examples enter the comparison, while this difference limits generalisation to the entire grid. Completing inference on the full grid would resolve that coverage gap.

The listening example illustrates one outcome and no formal listening study was conducted. A level-matched, blinded comparison could test whether the metric gains correspond to perceived clarity or quality. A follow-up model ablation could reconstruct each estimated magnitude with mixture and estimated phase under the model’s own transform settings. That design would distinguish the contribution of the model’s phase output from improvements in its magnitude estimate.

Reproducing the demonstration and final analysis

Run `python experiments/demo.py` with the bundled VoiceBank and DEMAND WAV files to generate audio and a score table at the three SNRs. Run `python experiments/analyze_final_results.py` to regenerate the paired tables, subgroup results and figures from the archived CSVs. Repository instructions describe the external data and checkpoint needed for the larger runs.

The corpus, noise recordings, pretrained model and metric implementations are credited in the references and repository. The full and matched archives provide the evidence for this report. AI assistance was used to help revise the analysis code, prepare the report and assemble the demonstration. The demonstration uses synthetic narration.

References and resources

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Code resources

MP-SENet · pystoi · PESQ Python wrapper